CHM2050 EXAM 2 — Ch 4.6–8.5 (Bonding · Composition · Stoichiometry · Gases) — FRONT

GLOBAL RULES (lose points otherwise): units on every answer + correct sig figs · show work · T in **K** · balance eqn FIRST · **exam counts /200 but has 225 pts → do easy problems first**, skip & return.

1. LEWIS STRUCTURES (4-step recipe)

1. **Count valence e⁻**: sum group e⁻; **+1 per (−) charge, −1 per (+) charge**. (SO_4^{2-} = 6+24+2 = 32)
2. **Skeleton**: central = lowest EN / largest / most valences (C, N, B, S, P, Xe, Cl w/ O). **H & F always peripheral**. Peripheral atoms bond only to central atoms.
3. Fill outer atoms first → then central; make multiple bonds until octets (H = 2).
4. **Formal charge check**: FC = VE − NBE − ½BE (= VE − lone-pair e⁻ − # bonds)

Pick best structure: (a) FCs minimized/zero (S=C=S not C=S=S) (b) no like-sign FC on neighbors (ONOO⁻) (c) (−) FC on the *more EN* atom (N_2O: N=N=O / N≡N–O) (d) least FC variation. Equivalent options → **resonance** (NO_3^-, O_3, CO_3^{2-}, benzene).

2nd-row atoms (C,N,O,F) NEVER exceed 8 e⁻. Period ≥3, groups 14–18 can be **hypervalent** (PCl_5, SF_6, XeF_4, SF_4). **Hypovalent**: B, Be, Al (BF_3 = 6 e⁻ on B, BeH_2 = 4). **Radicals**: odd e⁻ count (NO•, NO_2•, O_2^-•).

Bond order (Lewis) = avg # bonds over resonance structures = total bond pairs ÷ # equivalent bonds: O_3 = 1.5, NO_3^- = 4/3, CO_3^{2-} = 4/3, NO_2^- = 1.5, SO_4^{2-} (octet) = 1. Higher BO → shorter & stronger bond.

2. BOND TYPE & DIPOLES

ΔEN	bond
≤ 0.4	(non-polar) covalent
0.4 – 1.7	polar covalent
> 1.7 (metal + nonmetal)	ionic (e.g. YBr_3: 2.96−1.22 = 1.74 → ionic)
metal–metal	metallic (pooled e⁻ "sea")

μ = δ·e·R %-ionic = δ×100% = μ/(eR)×100% 1 D = 3.336×10⁻³⁰ C·m, e = 1.602×10⁻¹⁹ C, 1 Å = 10⁻¹⁰ m

HCl: μ=1.109 D, R=1.284 Å → δ = (1.109×3.336e−30)/(1.602e−19×1.284e−10) = 0.180 → **18.0 % ionic**

DIPOLE ARROW (IUPAC, used in THIS class): draw from δ⁻ (negative) → δ⁺ (positive). Textbook uses old +→− convention; **use the class convention!** e.g. H–I: arrow points from I to H.

Molecular dipole = **vector sum** of bond dipoles (CO_2 cancels → 0; H_2O bent → polar).

3. VSEPR (4.6) — steric # = atoms bonded + lone pairs on central

SN	e⁻ geom / hybrid	AX_mE_n → shape	angle
2	linear · sp	AX_2 linear (CO_2, BeH_2, C_2H_2)	180°
3	trig planar · sp^2	AX_3 trig planar (BF_3, SO_3, NO_3^-)	120°
		AX_2E bent (SO_2, O_3, NO_2^-)	<120°
4	tetrahedral · sp^3	AX_4 tetrahedral (CH_4, CF_4, SO_4^{2-})	109.5°
		AX_3E trig pyramidal (NH_3 107°, DMSO)	<109.5
		AX_2E_2 bent (H_2O 104.5°, OF_2)	<<109.5
5	trig bipyramid · sp^3d	AX_5 trig bipyramidal (PF_5, PCl_5)	90,120,180
		AX_4E **seesaw** (SF_4, $TeCl_4$)	<90,<120, <180
		AX_3E_2 T-shaped (ClF_3)	<90
		AX_2E_3 linear (XeF_2, I_3^-)	180°
6	octahedral · sp^3d^2	AX_6 octahedral (SF_6)	90°
		AX_5E square pyramidal (BrF_5)	<90°
		AX_4E_2 square planar (XeF_4)	90°

Space needed: lone pair > triple > double > single bond > single-e⁻ domain. LPs in trig-bipyramid go EQUATORIAL (only 2 neighbors at 90° vs 3 axial). In octahedral, 2 LPs go opposite (trans) → square planar. Distortions: SF_4 axial 173°, eq 106°. Double bond squeezes others (CH_2O: H–C–H <120°). NO_2^+ (180°) > NO_2• (134°) > NO_2^- (115°). **H_2S (92°) < H_2O (104.5°) < NH_3 (107°) < CH_4 (109.5°)**. Bigger ligands → bigger angle (OCl_2 > H_2O ≈ OF_2) — "ligand close packing".

Electron-pair geometry counts bonds + LPs; **molecular shape** names only atom positions.

POLARITY RULE: polar bonds + asymmetric shape = POLAR. **Nonpolar if all X identical AND shape is AX_2, AX_3, AX_4, AX_5, AX_6, AX_2E_3 (linear), AX_4E_2 (square planar)**. All other AX_mE_n are polar (bent, pyramidal, seesaw, T, sq-pyramidal). Different X's → usually polar.
e.g. XeF_4 nonpolar · **SF_4 polar** · CF_4 nonpolar · PF_5 nonpolar · SeF_4 polar · BeF_2, SiF_4 nonpolar.

Natta (line-dash-wedge) projection: line = in plane, wedge = toward you, dash = behind. Draw tetrahedral as 2 lines + 1 wedge + 1 dash; draw LPs.

4. VALENCE BOND THEORY & HYBRIDIZATION (5.1–5.3)

Bond = **overlap of 2 half-filled orbitals** (same phase/color), 1 e⁻ each → pair (Pauli). More overlap = stronger bond. Opposite phase → node → antibonding. Optimum bond length = overlap gain vs. nuclear repulsion.

σ bond: head-on, along bond axis, cylindrical (s–s, s–p, p–p end-on, hybrid–hybrid/–s). **π bond**: sideways p–p, density above/below axis, weaker than σ, nodal plane on axis.

Single = 1σ · Double = 1σ+1π · Triple = 1σ+2π (π's are perpendicular: p_x–p_x and p_y–p_y)

SN	hybrid	# HO	%s	leftover p (for π)	ex.
2	sp	2	50	2 p	C_2H_2, $BeCl_2$, CO_2 C
3	sp^2	3	33	1 p	C_2H_4, BF_3, benzene
4	sp^3	4	25	0	CH_4, C_2H_6, H_2O O, NH_3
5	sp^3d (d_{z^2})	5			PCl_5 (axial longer: 214 vs 202 pm)
6	sp^3d^2 (d_{z^2}, $d_{x^2-y^2}$)	6			SF_6; TM: d^2sp^3

Steps: promote (2s→2p) → hybridize (mix n AOs → n HOs, *orbital conservation*; HO energy = avg) → form bonds (energy gained > promotion cost). **Lone pairs count toward SN** (O in H_2O = sp^3).

"Describe bonds in C_2H_2 using VB" template (practice Q2): H–C≡C–H, linear, each C **sp**, 180°. Put molecule on **z-axis**. C–H: σ(C sp + H 1s). C–C: 1σ(sp+sp, head-on along z) + π($2p_x$+$2p_x$) + π(2p_y+2p_y). Sketch: two sp lobes overlapping on z; p_x lobes above/below (x), p_y lobes front/back. Total: 3σ + 2π.
C_2H_4: sp^2, 120°, C=C = σ(sp^2–sp^2) + π(2p_z–2p_z ⊥ plane); 5σ+1π. **C_2H_6**: sp^3, 7σ. **Oxirane C_2H_4O** (3-ring C–C–O, 2 H per C, 2 LP on O): **all three atoms sp^3** (strained ~60° ring).

5. MO THEORY (5.4) — LCAO

MOs spread over whole molecule. **n AOs → n MOs**. Bonding (constructive, +, e⁻ between nuclei, lower E); antibonding (*) (destructive, extra node, higher E). Antibonding raised MORE than bonding lowered → He_2, Be_2, Ne_2 don't exist. g = gerade (symmetric), u = ungerade.

BO = ½ (bonding e⁻ − antibonding e⁻)

Order — Li_2…N_2 (s–p mixing): σg2s < σ*u2s < **πu2p (×2) < σg2p** < π*g2p (×2) < σ*u2p
Order — O_2, F_2, Ne_2: σg2s < σ*u2s < **σg2p < πu2p (×2)** < π*g2p (×2) < σ*u2p

Fill bottom-up, Hund's rule in degenerate π's. **Unpaired e⁻ → paramagnetic**; all paired → diamagnetic. Total spin S = ½ × (# unpaired).

valence e⁻	species	BO	mag
2 / 4	Li_2 / Be_2	1 / 0	dia
6	B_2	1	**para**
8	C_2	2	dia
10	N_2, CO, NO^+, CN^-, **C_2^{2-}**	3	dia
11	N_2^-•, NO•, O_2^+•	2.5	para
12	O_2 (NO^-)	2	**para (2 unpaired)**
13	O_2^-• superoxide	1.5	para, S=½
14	O_2^{2-} peroxide, F_2	1	dia
16	Ne_2	0	–

Practice Q3 (O_2 family): O_2: $(\sigma g2s)^2(\sigma^*u2s)^2(\sigma g2p)^2(\pi u2p)^4(\pi^*g2p)^2$ BO 2 → **120.8 pm**; O_2^+• $(\pi^*)^1$ BO 2.5 → **111.6**; O_2^-• $(\pi^*)^3$ BO 1.5 → **135.0**; O_2^{2-} $(\pi^*)^4$ BO 1 → **145.3 pm**. (higher BO = shorter)
H_2 BO 1; H_2^+• ½; H_2^-• ½; He_2^+• ½; He_2 0.

Heteronuclear (HF, NO, CO): more EN atom's AOs are LOWER in energy; bonding MO looks more like (closer in E to) the EN atom's AO → e⁻ pulled toward it. AOs that don't overlap = **non-bonding MOs** (= lone pairs). Big energy mismatch → ionic.

VB vs MO: VB = localized 2-center bonds + LPs (basis of Lewis); MO = delocalized, includes antibonding, explains O_2 paramagnetism & fractional BO.

6. "MOLECULES OF THE DAY" LEWIS QUICK-REF

O_3: bent 117°, resonance, BO 1.5, central O FC +1. O_3^-•: bent radical. NO_2^-: bent, BO 1.5 (ambidentate). NO_3^-: trig planar, N FC +1, BO 4/3. $ONOO^-$ peroxynitrite: O=N–O–O⁻ (avoid like-sign FCs). N_2O: N=N=O ↔ N≡N–O. NO•, NO_2•: radicals (odd e⁻). SO_2: bent. SO_3: trig planar. SO_4^{2-}, SO_3^{2-} (pyramidal), $S_2O_3^{2-}$ (one O→S). **DMSO $(CH_3)_2S$=O: S central, 2 CH_3 + O + 1 LP → trig pyramidal at S, sp^3; M = 78.13 g/mol**. Alkanes C_nH_{2n+2} all sp^3; ethene sp^2. Peroxo = O–O single bond.

MO FILLING DIAGRAM — O_2 (draw this on exam)

MO	O_2 (12)	N_2 (10) order
σ*u2p	__	σ*u2p __
π*g2p ×2	↑_ ↑_ (2 unpaired → para)	π*g2p __ __
πu2p ×2	↑↓ ↑↓	**σg2p** ↑↓ (above π!)
σg2p	↑↓	πu2p ↑↓ ↑↓
σ*u2s	↑↓	σ*u2s ↑↓
σg2s	↑↓	σg2s ↑↓

AOs drawn on each side (2s low, 2p high), MOs in middle, dashed lines connect. O_2: BO = ½(8−4) = 2.

QUICK ANSWERS (lecture/LC questions)

- **EN order**: K < Be < Sn < Se < S < Cl. Trend: EN ↑ across a period, ↓ down a group; F 3.98 highest. (Exam gives EN table.)
- **Central atoms**: H_2O→O, NCl_3→N, SO_2→S, HCN→C, $HClO_3$→Cl, XeF_4→Xe, PCl_5→P. H: 1 bond, O: 2, N: 3, C: 4, B: 3.
- **FC on N in NO_3^- = +1** (each single-bonded O −1). Best $ONOO^-$ = O=N–O–O⁻ (FC only on terminal O).
- BF_3: B 6 e⁻ (no double bond, FC 0). SO_4^{2-} octet version: S +2, each O −1 (expanded S=O versions also drawn). SF_6: S 12 e⁻.
- KF potential curve: R_e = equilibrium bond length; ΔE_d = energy to separate into neutral atoms; V(R_e) involves Coulomb attraction + repulsion (not Coulomb only).
- AX_nE_2 can be nonpolar only for **n = 4** (sq planar). AX_nE (n=2–5): **none** can be nonpolar. Polar among BeF_2, SiF_4, SeF_4, XeF_4 → **SeF_4**.
- Wrong hybridization statement: "180° w/ two ligands can *always* be described by sp" (e.g. XeF_2 is AX_2E_3 → sp^3d).
- BO: H_2^+• ½ · H_2^-• ½ · N_2^-• 2.5 · C_2^{2-} 3 (K_2C_2 **diamagnetic**) · O_2^-• total spin ½ · NO_3^- N–O 4/3.
- Count σ/π: every bond has 1σ; extra bonds are π. HCN: 2σ + 2π. CO_2: 2σ + 2π (C sp, O sp^2).

BACK — Composition · Solutions · Reactions · Stoichiometry · Gases · Safety · Playbook

7. COMPOSITION (6.1–6.2)

n = m / $\mathcal{M}$ · N = n·N_A (6.022×10²³) · 1 Da = 1 g/mol · **mass-% (X) = (# X × $\mathcal{M}_X$) / $\mathcal{M}_{compound}$ × 100**

EMPIRICAL FORMULA from %: (1) assume **100 g** → % = grams (2) ÷ atomic mass → mol (3) **÷ smallest** (4) fix decimals: x.5→×2, x.33/x.67→×3, x.25/x.75→×4, x.2→×5 (5) write in Hill order (C, H, then alphabetical). **MOLECULAR FORMULA**: k = $\mathcal{M}_{molecular}$ / $\mathcal{M}_{empirical}$ (round to integer) → multiply all subscripts by k.

40.0% C, 6.7% H, 53.3% O → 3.33 : 6.65 : 3.33 → **CH_2O**. If $\mathcal{M}$≈60 → k=2 → $C_2H_4O_2$.
Dye 75.95% C, 17.72% N, 6.33% H, $\mathcal{M}$≈240: 6.323 : 1.265 : 6.280 → C_5H_5N (79.10) → k = 3.03 → **$C_{15}H_{15}N_3$**.
Hematite 34.97 g Fe, 15.00 g O → 0.6262 : 0.9375 → 1 : 1.497 → ×2 → Fe_2O_3. Empirical CH, $\mathcal{M}$ 78.11 → C_6H_6.

8. SOLUTIONS (6.3–6.4)

Molarity M = n_{solute} / $V_{solution}$(L)	T-dependent (volume changes)
Molality b = n_{solute} / $m_{solvent}$(kg)	mass only, T-independent, more precise
Mole fraction χ = n_i/n_{total}, Σχ=1	mass-% = m_i/m_{tot}×100; ppm = ×10⁶
Dilution: $c_1V_1 = c_2V_2$ (moles of solute conserved) · $n_{ions} = i \cdot n_{formula}$ (NaCl i=2, $CaCl_2$ i=3, $Fe(NO_3)_3$ i=4)	
Dilute aqueous: **M ≈ b** (d_{water}≈1 kg/L), M ≈ $d_{solvent}$·b · χ ↔ b: b = χ / [(1−χ)·$\mathcal{M}_{solvent}$(kg/mol)]	

w/w% + density → b or M: take 100 g solution; solute g = %, solvent g = 100 − %; V = 100 g / d.

64.0% H_2SO_4, d=1.5421: 64.0/98.079 = 0.6525 mol; 36.0 g water → **b = 18.1 mol/kg**; V = 64.85 mL → M = 10.1 M.
Saline 0.9% NaCl ≈ 9 g/L → **0.154 M NaCl = 0.308 M ions**. 1 L + 1 L water → 0.077 M.
Q8: V_1 = (0.154 M)(500. mL)/(1.00 M) = **77.0 mL** stock, fill to 500. mL with DI water in volumetric flask.

Making a MOLAR soln: weigh solute → dissolve in some solvent in **volumetric flask** (below mark) → rinse all solute in → fill **to the mark** → mix. **MOLAL soln**: weigh solute AND weigh solvent on analytical balance → mix (avoid evaporation).

HYDRATE TRAP (Q9): 1 L of 0.1594 m $Fe(NO_3)_3$ from $Fe(NO_3)_3 \cdot 9H_2O$ ($\mathcal{M}$ = 241.86 + 9×18.015 = **403.99 g/mol**). Need ≈1 kg water → 0.1594 mol → **64.40 g hydrate**. The $9H_2O$ joins the solvent: 9×0.1594×18.015 = 25.84 g → add only **974.2 g DI water**. Weigh both, mix, dissolve. (≈1 L since dilute.)

9. REACTIONS (7.1–7.2) — always take INVENTORY

Inventory = list every species actually in each beaker (incl. H_2O(l)). Strong electrolytes → ions only. Then ask: which new combination is insoluble / acid+base / gas / redox? Remove **spectator ions** → **net ionic equation** (balance atoms AND charge).

SOLUBLE	exceptions (insoluble)
Group 1 (Li^+,Na^+,K^+…) & NH_4^+ — always	–
NO_3^-, acetate — always	–
Cl^-, Br^-, I^-	**Ag^+, Pb^{2+}, Hg_2^{2+}** ($AgCl$, PbI_2, Hg_2Cl_2)
SO_4^{2-}	Ca^{2+}, Sr^{2+}, Ba^{2+}, Pb^{2+}, Ag^+
INSOLUBLE	**exceptions (soluble)**
OH^- (TM & Al^{3+} insoluble; Ca,Sr,Ba slightly)	Group 1, NH_4^+
S^{2-} (TM, Pb, As, Sb, Bi), CO_3^{2-}, PO_4^{3-}, CrO_4^{2-}, F^- (Ba, Mg, Pb)	Group 1, NH_4^+
Most Ag^+ salts insoluble (except $AgNO_3$, Ag acetate)	

Q10 KI + $Pb(NO_3)_2$: (a) KI(aq): K^+, I^-, H_2O · $Pb(NO_3)_2$(aq): Pb^{2+}, NO_3^-, H_2O (b) **precipitation** (yellow) (c) **Pb^{2+}(aq) + 2 I^-(aq) → PbI_2(s)**; K^+, NO_3^- spectators. (0.0050 mol each → I^- limiting → 0.0025 mol PbI_2.) Driving force: Coulomb attraction (ion–ion > ion–dipole).

Why water dissolves ions: polar, bent, dipole → **ion–dipole** forces (O end → cations, H end → anions). Solubility = balance of lattice ion–ion vs ion–dipole.

Acid/Base (Brønsted–Lowry)

Acid = H^+ donor, base = H^+ acceptor; acid → conjugate base, base → conjugate acid. Arrhenius: ↑H_3O^+ / ↑OH^- in water. **Amphoteric**: H_2O, HCO_3^-, HSO_4^-, HPO_4^{2-}, $[Mg(OH_2)_5(OH)]^+$ (CA = $[Mg(OH_2)_6]^{2+}$, CB = $[Mg(OH_2)_4(OH)_2]$).
Strong acids (100% ionized, sheet): $H_2S_2O_7$, HI, HBr, HCl, H_2SO_4 (1st H^+), $HClO_4$, HNO_3 (+$HClO_3$). **Strong bases**: OH^-, O^{2-}, S^{2-}, NH_2^- (metal hydroxides/oxides). **Weak**: HF, CH_3COOH, H_3PO_4, H_2CO_3, NH_4^+, H_2S, HCN / NH_3, CO_3^{2-}, CN^-, F^-, acetate.
Inventories: strong HA: H_3O^+, A^-, H_2O (no HA). Weak HA: HA, A^-(little), H_3O^+(little), H_2O. NaOH: Na^+, OH^-, H_2O. **H_2SO_4(aq): H_3O^+, HSO_4^-, H_2O** (little SO_4^{2-}). K_2O: O^{2-} + H_2O → 2OH^- (no O^{2-} in water).
Net ionic: strong+strong: H_3O^+ + OH^- → 2H_2O. Weak acid + OH^-: CH_3COOH + OH^- → CH_3COO^- + H_2O (write weak acid intact). **Gas-forming**: CO_3^{2-} + 2H_3O^+ → H_2CO_3 + 2H_2O → **CO_2(g) + 3H_2O** (also S^{2-}→H_2S, SO_3^{2-}→SO_2).
Acid-strength trends: → across row EN↑ → stronger (CH_4<NH_3<H_2O<HF). ↓ down group size↑ → stronger (**HF weakest, HI strongest**; lower charge density = stabler anion). More O's → stronger (HNO_3 > HNO_2 > HNO). Nonmetal oxides acidic (SO_3 > SO_2), metal oxides basic.

Redox & oxidation numbers

OIL RIG: Oxidation = Loses e^- (ON ↑); Reduction = Gains e^- (ON ↓). **Oxidizer gets reduced; reducer gets oxidized.**
ON rules: element = 0 · monatomic ion = charge · Σ = overall charge · **F = −1** · O = −2 (peroxide −1, with F +2) · Group 1 +1, Group 2 +2, Al +3 · H +1 (−1 with metals) · halogens −1 (unless with O). From Lewis: give ALL bond e^- to more EN atom (split if equal). **ON ≠ formal charge**. Ex: NO_3^- N=+5; CH_3COOH: CH_3 C = −3, COOH C = +3. Thermite Fe_2O_3 + 2Al → 2Fe + Al_2O_3: Fe +3→0 (oxidizer), Al 0→+3 (reducer). Combustion = redox (C oxidized, O reduced), exothermic ΔH < 0.

10. STOICHIOMETRY (7.3–7.5)

ROAD MAP: g A ÷$\mathcal{M}_A$→ mol A ×(coef B/coef A)→ mol B ×$\mathcal{M}_B$→ g B. Solutions: mol = M×V(L). Gases: mol = PV/RT.
LIMITING REACTANT: mol of each ÷ its coefficient → smallest = limiting (or: compute product from each; least product wins). Theoretical yield from limiting.
Excess left = given − used.
%-yield = actual / theoretical × 100%

Balancing: balance metals/other first, then H, then O last; smallest whole numbers; check charge for ionic eqns.

Q7: **$2NH_3$ + 3CuO → N_2 + 3Cu + 3H_2O**. 18.1 g/17.031 = 1.063 mol NH_3 (÷2 = 0.531); 90.4/79.545 = 1.136 mol CuO (÷3 = 0.379) → **CuO limiting**. N_2 theo = 1.136/3 × 28.014 = 10.61 g; %-yield = 6.63/10.61 = **62.5%**.
Mg + 2HCl → $MgCl_2$ + H_2: 15.8 g Mg (0.650 mol), 14.6 g HCl (0.400 mol ÷2 = 0.200) → HCl limiting → 0.200×95.21 = 19.0 g; 16.3 g actual → 85.8%.
P_4 + 6Cl_2 → 4PCl_3: 31.3 g Cl_2 needs 9.12 g P_4 → 45.0 − 9.1 = **35.9 g P_4 excess**.

Titration

Titrant (known M) added from buret until **equivalence point** (stoichiometric) ≈ **end point** (indicator color change). **$M_{titrant}V_{titrant}$ → mol → ×ratio → mol analyte ÷ $V_{analyte}$**. Primary standard: very pure, stable, no hydrate water, high $\mathcal{M}$.

Q11: 2Cl^- + Hg^{2+} → $HgCl_2$(s). n Hg^{2+} = 1.46×10⁻³ L × 8.25×10⁻⁴ M = 1.205×10⁻⁶ mol → ×2 = 2.41×10⁻⁶ mol Cl^- ÷ 0.25×10⁻³ L = **9.6×10⁻³ M** (2 sf from 0.25 mL).
Don't forget the 2:1 ratio!
0.500 M NaOH vs 500. mL 0.100 M HAc (1:1) → 0.0500 mol → 0.100 L = 100. mL. $Ba(OH)_2$ + H_2SO_4 → $BaSO_4$(s) + 2H_2O (acid/base + precipitation).

11. GASES (8.1–8.5)

PV = nRT · R = **0.08206 L·atm/(mol·K)** = 8.314 J/(mol·K) = 62.36 L·Torr/(mol·K) · **T(K) = °C + 273.15**
Combined: $P_1V_1/(n_1T_1) = P_2V_2/(n_2T_2)$ → cancel what's constant (Boyle $P_1V_1=P_2V_2$; Charles V/T; Gay-Lussac P/T; Avogadro V/n)

1 atm = 760 mmHg = 760 Torr = 101.325 kPa = 1.01325 bar = 14.7 psi; 1 L·atm = 101.325 J. Barometer P = ρgΔh (Hg 13.56 g/mL). **STP** 0 °C, 1 atm → **22.414 L/mol**.
Density d = P$\mathcal{M}$/(RT) · **Molar mass $\mathcal{M}$ = mRT/(PV) = dRT/P**
Dalton: $P_{tot} = \Sigma P_i$, $P_i = \chi_i P_{tot}$. Collected over water: $P_{gas} = P_{atm} - P_{H_2O}(T)$. Vapor pressure ↑ with T; = P_{atm} at normal b.p.
Graham (8.4): $rate_1/rate_2 = \sqrt{(\mathcal{M}_2/\mathcal{M}_1)}$; $time_1/time_2 = \sqrt{(\mathcal{M}_1/\mathcal{M}_2)}$ (lighter = faster).
KMT (8.5): $\langle E_{kin} \rangle$ = 3/2 k_BT (per particle) = 3/2 RT (per mol) — depends ONLY on T. **$u_{rms} = \sqrt{(3RT/\mathcal{M})}$**, $\langle v \rangle = \sqrt{(8RT/\pi\mathcal{M})}$, $v_{mp} = \sqrt{(2RT/\mathcal{M})}$ **$\mathcal{M}$ in kg/mol, R = 8.314!**
$v_{mp} < \langle v \rangle < u_{rms}$. Postulates: point particles, no forces, elastic collisions, random motion. **Real gases deviate at high P, low T**; Z = PV/nRT (=1 ideal); He most ideal.
Gas stoich: same T,P → **volume ratio = mole ratio**. $2C_2H_2$ + $5O_2$ → $4CO_2$ + $2H_2O$: 9340 L C_2H_2 × 5/2 = 23 350 L O_2 ÷ 7000 L = 3.34 → **4 tanks**.

Q12: read graph at 32 °C → **P_{H_2O} ≈ 0.047 atm**; n = PV/RT = 0.047×1.00/(0.08206×305.15) = 1.9×10⁻³ mol × 18.016 = 0.034 g ÷ 995.1 g/L → **0.034 mL**.
Q13: $V_2 = V_1(P_1/P_2)(T_2/T_1)$ = 1.00 L (760/770) (330.15/300.15) = **1.09 L** → **expands**: T rises 10% but P only 1.3% → T effect wins.

12. SAFETY MINUTE (asked on exam!)

Autoclave hazards: scalding high-P steam (~30 psi) burns; superheated liquids boil over/splash if removed too soon; steam corrosion → vessel failure/explosion (annual inspection). **PPE**: eye protection/face shield, lab coat/apron, **heat-insulating gloves** + training. **Pressurized devices**: never open before pressure released (MagLab 2015 death). **NFPA diamond**: blue = health, red = flammability, yellow = instability/reactivity, white = special (W, OX); 0–4 (4 worst). **SOP** = procedures, hazards, controls, training documentation. **SDS** 16 sections (1 ID, 2 hazards, 4 first aid, 5 fire, 7 handling/storage, 8 exposure/PPE, 11 toxicology). **Pyrophorics** (t-BuLi, Sangji UCLA 2009: no lab coat, syringe at 50 mL scale): syringe <50 mL, **cannula ≥50 mL**, septa (≤~10 punctures), flame-resistant lab coat. **Complacency**: stay alert in routine work.

13. EXAM PLAYBOOK (by question type)

1. **Lewis + VSEPR + polar?** count e^- → skeleton → octets/FC → SN → e^- geom → shape (LP placement) → Natta drawing → dipole vectors → polar/nonpolar. State angles & distortions.
2. **VB description**: Lewis → hybridization per atom (SN) → draw xyz axes, bond axis = z → label every bond σ/π + which orbitals overlap (sp–1s, sp–sp, p–p) → sketch overlap.
3. **MO / bond order**: count valence e^- (adjust charge) → choose ordering (≤N_2 vs O_2/F_2) → fill → BO → para/dia → rank bond lengths (high BO = short).
4. **Molar mass**: Σ(atoms × atomic mass) with 4–5 sf; show each term.
5. **Empirical/molecular**: 100 g → mol → ÷ smallest → integers → k = $\mathcal{M}/\mathcal{M}_{emp}$.
6. **Limiting/yield**: balance → mol → ÷ coef → smallest limits → theo yield → %.
7. **Dilution**: $c_1V_1 = c_2V_2$ + describe flask procedure.
8. **"Explain how to make…"**: give numbers (g solute, g or mL solvent), glassware, steps; hydrates!
9. **Mixing two solutions**: (a) inventory each (b) classify rxn type (c) net ionic + phases + spectators.
10. **Titration**: mol titrant → ratio → mol analyte → ÷ V.
11. **Gas**: list P,V,n,T with units; convert to atm, L, K; pick law; predict qualitatively first (sanity check).
12. **Safety**: 2 specific hazards + 3 specific PPE items.

Sig-fig reminders: ×/÷ → fewest sf; +/− → fewest decimals; exact numbers (coefficients, 273.15 def., 1000 mL/L) unlimited. "500." has 3 sf. Keep extra digits until the end.